INCH-POUND
MS18301C(SH)
w/AMENDMENT 1
19 October 2021
SUPERSEDING
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08 November 2016

MILITARY SPECIFICATION SHEET

HIGH PRESSURE DRAIN ORIFICE PLATES AND STRAINERS

This specification sheet is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

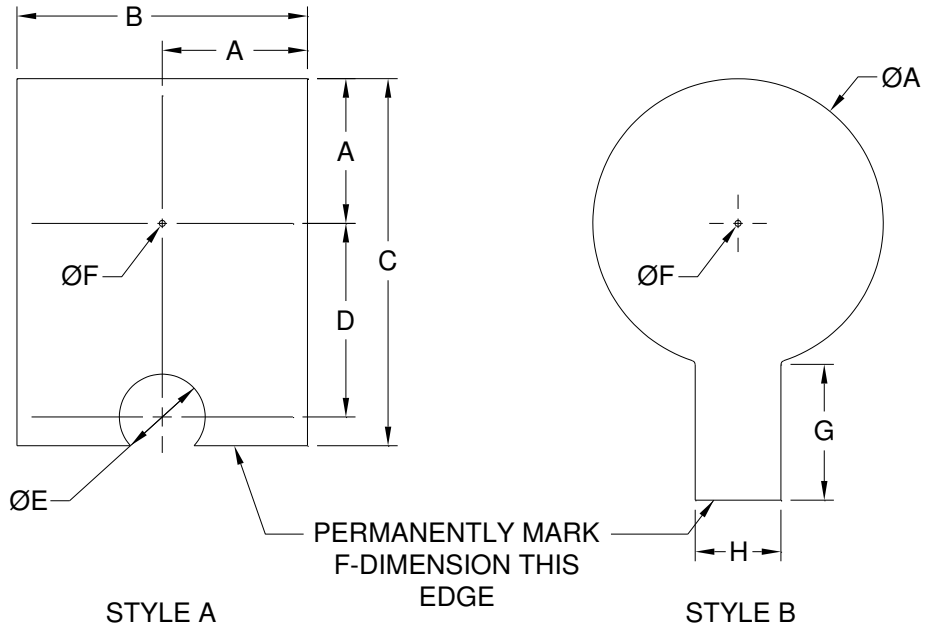
The requirements for acquiring the product described herein shall consist of this specification sheet.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

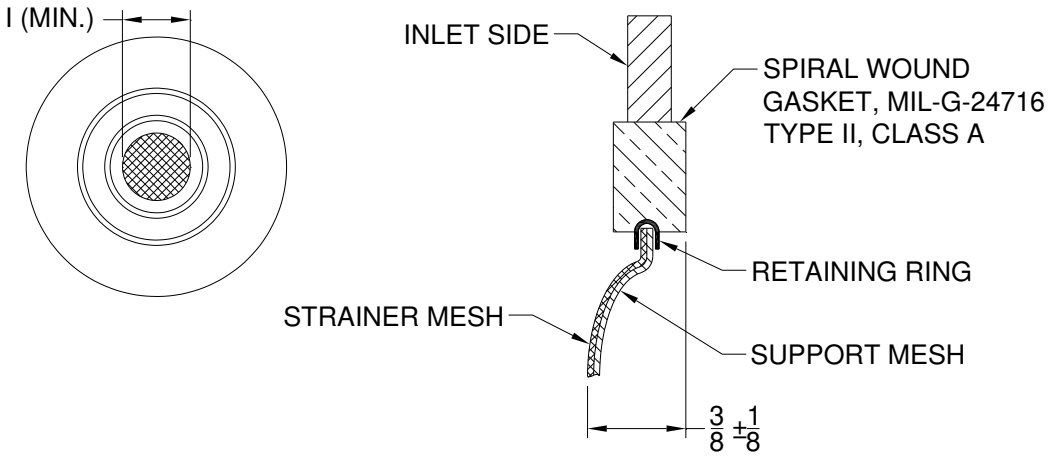
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ORIFICE PLATE
DASH NO. 01, 02, 03, 04, 05, 06, 15, 16, 17, & 18



DOMED STRAINER/GASKET
DASH NO. 07, 08, 09, 10, 11, 12, 13, & 14

FIGURE 1. Orifice plate and domed strainer/gasket.

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TABLE I. Orifice dimensions (inches).

Dash No.	Series	IPS	A ± 1/64	B ± 1/64	C ± 1/32	D ± 1/64	E ± 1/64	F ± 0.001	G ± 1/64	H ± 1/64	I (min.)
01	1500	½	1¼	2½	2 15/16	1⅝	⅞	0.020	1⅛	¾	--
02	1500	¾	1⅜	2¾	3 3/16	1¾	⅞	0.020	1 3/16	¾	--
03	600	½	1 1/16	2⅛	2 7/16	1 5/16	⅝	0.031	13/16	¾	--
04	600	¾	1 5/16	2⅝	3	1⅝	¾	0.031	1	¾	--
05	150	½	15/16	1⅞	2 3/16	1 3/16	⅝	0.031	13/16	¾	--
06	150	¾	1⅛	2¼	2 9/16	1⅜	⅝	0.031	13/16	¾	--
07	1500	½	--	--	--	--	--	--	--	--	½
08	1500	¾	--	--	--	--	--	--	--	--	¾
09	600	½	--	--	--	--	--	--	--	--	½
10	600	¾	--	--	--	--	--	--	--	--	¾
11	150	½	--	--	--	--	--	--	--	--	½
12	150	¾	--	--	--	--	--	--	--	--	¾
13	1500	1	--	--	--	--	--	--	--	--	1
14	600	1	--	--	--	--	--	--	--	--	1
15	1500	½	1¼	2½	2 15/16	1⅝	⅞	0.036	1⅛	¾	--
16	1500	¾	1⅜	2¾	3 3/16	1¾	⅞	0.036	1 3/16	¾	--
17	1500	1	1 9/16	3⅛	3⅜	2	1	0.020	1⅜	¾	--
18	600	1	1 7/16	2⅞	3¼	1¾	¾	0.031	1	¾	--

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REQUIREMENTS:

1. Orifice plates shall be constructed of ¼-inch thick, Type 304 corrosion-resistant steel plate. Surface finish in gasket contact area shall not exceed 500 roughness height rating (RHR).

2. Orifice plates, dash numbers 01, 02, 03, 04, 05, 06, 15, 16, 17, and 18, may be constructed in either Style A or Style B at manufacturer's option.

3. Orifice plate material shall be certified. Mill certification of materials will be accepted in lieu of performing chemical and physical tests.

4. Domed strainer/gasket dash numbers 07, 08, 09, 10, 11, 12, 13, and 14:

   a. Strainer portion of strainer/gasket assembly shall be constructed of Type 18-8 corrosion resisting steel alloy.
      (1) Strainer mesh shall be 60 by 60 mesh with 0.0075-inch diameter wire.
      (2) Support mesh shall be 16 by 16 mesh with 0.018-inch diameter wire.
      (3) Retaining ring shall be crimped to hold strainer and support mesh together.
   b. Strainer shall be positively retained as an integral part of a spiral wound gasket in accordance with MIL-G-24716, Type II, Class A: series (as specified by dash number).
      (1) Gasket compression characteristics shall not be affected by the strainer.
      (2) Metal outer ring shall be permanently marked "inlet" on the strainer mesh side.

5. Pieces detailed on this military specification sheet are designed for use in conjunction with gaskets specified as MIL-G-24716, Type II, Class A and flanges specified in ASME B16.5.

6. Sharp edges shall be removed.

7. All dimensions are in inches.

8. Dimension F shall be inspected in accordance with the requirements of MIL-STD-1916, Verification Level VII.

9. For design feature purposes, this specification sheet takes precedence over procurement documents referenced herein.

10. The issues of these documents are those cited in the solicitation or contract:

MIL-STD-1916, DoD Preferred Methods for Acceptance of Product

MIL-G-24716, Gaskets, Metallic-Flexible Graphite, Spiral Wound

(Copies of these documents are available online at https://quicksearch.dla.mil.)

ASME B16.5, Pipe Flanges and Flanged Fittings: NPS ½ through NPS 24 Metric/Inch Standard

(Copies of this document are available online at www.asme.org.)

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Amendment notations. The margins of this specification sheet are marked with vertical lines to indicated modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Preparing activity:
Navy – SH
(Project 4730-2021-044)

Review activity:
DLA – CC

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